Statistical Analysis of Economic Growth and Its Determinants in India

An Empirical Investigation of World Bank WDI Macroeconomic Indicators (1991–2025, n = 35)

Academic Subject Area:	Applied Macroeconomic Statistics & Econometrics
Target Economy:	Republic of India (IND)
Data Source:	World Bank World Development Indicators (WDI)
Temporal Horizon:	1991–2025 (35 Continuous Annual Observations)
Dependent Variable:	Real GDP Growth (annual percentage change)
Explanatory Determinants:	Inflation (CPI), FDI Net Inflows, Exports, Gross Capital Formation, Unemployment
Candidate / Student Name:	_____
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Faculty / Department:	Department of Economics & Applied Statistics
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Submission Period:	Academic Year 2025–2026

Notice: *This academic research assignment adheres to rigorous empirical standards. All statistical models, descriptive metrics, probability distributions, regression equations, and econometric diagnostics are derived dynamically from official World Bank WDI data without software promotion or marketing content.*

Table of Contents

Abstract **Page 3**

1. Introduction **Page 3**

1.1 Research Questions Page 3

1.2 Research Objectives Page 4

1.3 Scope and Delimitations of the Study Page 4

2. Dataset and Variables **Page 4**

2.1 Data Source and Retrieval Protocol Page 4

2.2 Study Period and Unit of Observation Page 4

2.3 Variable Definitions and Measurement Units Page 5

2.4 Conceptual Framework Page 5

3. Data Preparation and Quality Assessment **Page 5**

3.1 Data Collection and Assembly Protocol Page 5

3.2 Missing Value Audit and Duplicate Checks Page 6

3.3 Observation Counts Across Empirical Analyses Page 6

3.4 Data Integrity and Measurement Interpretation Page 6

4. Descriptive Statistics and Exploratory Data Analysis **Page 7**

4.1 Measures of Central Tendency Page 7

4.2 Measures of Dispersion and Variability Page 7

4.3 Comprehensive Numerical Descriptive Statistics Page 7

4.4 Annual GDP Growth Trajectory and Historical Shocks Page 8

4.5 Trends in Explanatory Macroeconomic Indicators Page 8

4.6 Quartiles, Interquartile Range, and Relative Dispersion Page 9

4.7 Skewness, Kurtosis, and Distribution Shapes Page 9

4.8 Outlier Screening via Tukey's Interquartile Range Page 9

4.9 Synthesis of Descriptive Findings Page 10

5. Probability Distribution Analysis **Page 10**

5.1 Grouped Frequency Distributions Page 10

5.2 Empirical Histograms and Fitted Normal Curves Page 10

5.3 Normal Quantile-Quantile (Q-Q) Plots and Departures Page 11

5.4 Shapiro-Wilk Statistical Normality Test Page 11

5.5 Anderson-Darling Goodness-of-Fit Test Page 11

5.6 Comparative Distributional Assessment Page 12

5.7 Empirical Probability Calculations and Applicability Page 12

6. Correlation Analysis **Page 12**

6.1 Pearson Product-Moment Linear Correlation Page 12

6.2 Spearman Rank Monotonic Correlation Page 13

6.3 GDP Growth and Explanatory Variable Associations Page 13

6.4 Inter-Predictor Correlation and Collinearity Screening Page 13

6.5 Synthesis of Bivariate Correlation Findings Page 14

7. Simple Linear Regression Analysis **Page 14**

7.1 to 7.5 Individual Bivariate Models (Inflation, FDI, Exports, GCF, Unemployment) Page 14

7.6 Comparative Synthesis of Bivariate Simple Models Page 15

8. Multiple Linear Regression Analysis **Page 15**

8.1 Multivariate Model Specification and Estimation Page 15

8.2 Multiple Regression Empirical Estimation Results Page 15

8.3 Individual Partial Coefficient Interpretations Page 16

8.4 Overall Goodness-of-Fit and ANOVA F-Test Evaluation Page 16
8.5 Heteroskedasticity-Consistent (HC3) Robust Inference Page 16
9. Regression Diagnostics and Model Validation Page 17
9.1 Linearity and Residual Dispersion Page 17
9.2 Residual Normality Evaluation and Q-Q Diagnostics Page 17
9.3 Breusch-Pagan Lagrange Multiplier Test Page 17
9.4 Durbin-Watson First-Order Autocorrelation Screening Page 18
9.5 Variance Inflation Factors (VIF) Multicollinearity Assessment Page 18
9.6 Cook's Distance and Point Influence Screening Page 18
9.7 Diagnostic Summary Table and Assessment Page 18
10. Discussion of Results Page 19
11. Conclusion and Policy Implications Page 19
References and Data Source Page 20
Appendices A through I Page 21

List of Tables and List of Figures

Table 1: Dataset Summary & Verification Parameters
Table 2: Definition & Specification of Variables
Table 2B: Observation Counts Across Analyses
Table 3: Comprehensive Descriptive Statistics (n=35)
Table 4: Annual Real GDP Growth Series (1991–2025)
Table 5: Quartiles, IQR, and Relative Dispersion
Table 6: Grouped Frequency Distribution of GDP Growth
Table 7: Outlier Screening via Tukey's Fences
Table 8: Shapiro-Wilk & Anderson-Darling Tests
Figure 3: Multi-Panel Boxplots & Outliers
Figure 5: Normal Q-Q Plots across Series
Figure 7: Spearman Rank Heatmap
Figure 9: Actual vs Fitted GDP Growth
Figure 11: Normal Q-Q Plot of Residuals
Figure 13: VIF Multicollinearity Bar Chart
Table 9: Empirical Macroeconomic Probabilities
Table 10: Linear vs Monotonic Correlation with GDP Growth
Table 11: Five Bivariate Simple Linear Regression Models
Table 12: Multiple Linear Regression (OLS vs HC3)
Table 13: Variance Inflation Factors (VIF)
Table 14: Comprehensive Econometric Diagnostics Summary
Table 15: Formal Statistical Hypothesis Testing Summary
Figure 1: GDP Growth Trajectory (1991–2025)
Figure 2: Multi-Panel Macroeconomic Trends
Figure 4: Histograms with Fitted Gaussian Curves
Figure 6: Pearson Correlation Matrix Heatmap
Figure 8: Fitted Bivariate Simple Regression Panel
Figure 10: Regression Residuals vs Fitted Values
Figure 12: Empirical Residual Distribution
Figure 14: Cook's Distance Stem Plot with 4/n Line

Abstract

Background and Problem Formulation: Economic growth serves as the fundamental barometer of macroeconomic health, living standard advancement, and productive capacity expansion in emerging economies. Following the watershed structural reforms initiated during the 1991 balance-of-payments crisis, India embarked upon an ambitious trajectory of market liberalization, trade integration, and external capital absorption. Nonetheless, the empirical relationships linking economic growth to key macroeconomic drivers—namely Inflation (Consumer Price Index annual %), Foreign Direct Investment (FDI Net Inflows % of GDP), Exports of Goods and Services (% of GDP), Gross Capital Formation (% of GDP), and Unemployment (% of total labor force)—remain characterized by theoretical tensions, non-linearities, and vulnerability to severe structural shocks.

Research Objectives and Methodological Framework: This academic research assignment executes a rigorous, multi-tiered statistical investigation into India's annual real GDP growth and its five core explanatory determinants. Utilizing official World Bank World Development Indicators (WDI) spanning the verified 35-year post-reform epoch (1991–2025, exactly n = 35 continuous annual observations, zero missing values, zero synthetic imputations), the empirical architecture progresses systematically from univariate parametric and non-parametric exploratory data analysis, frequency distributions, and theoretical probability fitting to bivariate linear (Pearson) and monotonic rank (Spearman) correlations, five simple ordinary least squares (OLS) regressions, a comprehensive multiple linear regression model, Davidson-MacKinnon HC3 heteroskedasticity-consistent robust inference, and an exhaustive battery of econometric specification diagnostics.

Principal Empirical Findings: Over the 35-year timeframe, India registered a mean annual real GDP growth rate of 6.14% (SD = 2.80%), ranging from a pandemic-induced trough of -5.78% (2020) to a peak expansion of 9.69% (2010). Normality testing at $\alpha = 0.05$ demonstrates that while Gross Capital Formation (Shapiro-Wilk W = 0.9691, p = 0.4193; Q-Q fit r = 0.9896) and FDI Net Inflows (W = 0.9431, p = 0.0700) adhere to Gaussian expectations, GDP Growth (W = 0.7735, p < 0.0001), Inflation (W = 0.9238, p = 0.0173), Exports (W = 0.9080, p = 0.0065), Unemployment (W = 0.6559, p < 0.0001), and regression residuals (W = 0.9006, p = 0.0041) deviate significantly from normality. Crucially, non-parametric Spearman rank correlation reveals statistically significant monotonic associations between GDP Growth and both Gross Capital Formation (rho = 0.4347, p = 0.0090) and Unemployment (rho = -0.3754, p = 0.0264), contrasting with their non-significant linear Pearson counterparts (r = 0.3180, p = 0.0626 and r = -0.2095, p = 0.2270). In multiple OLS estimation ($R^2 = 0.2840$, Adjusted $R^2 = 0.1605$, $F(5, 29) = 2.3005$, p = 0.0707), conventional OLS standard errors indicate nominal significance for FDI (p = 0.0488) and Gross Capital Formation (p = 0.0401). However, under HC3 robust covariance estimation, standard errors expand substantially (+41.8% for FDI, +38.5% for GCF), rendering ALL five predictors statistically non-significant at $\alpha = 0.05$.

Conclusions and Methodological Caveats: Diagnostic screening confirms absence of severe first-order serial correlation (Durbin-Watson = 1.7435) and heteroskedasticity (Breusch-Pagan LM = 8.6985, p = 0.1217). However, moderate multicollinearity is present (Exports VIF = 5.70), and Cook's distance identifies two highly influential points (1991, D = 0.1786 and 2020, D = 0.6139) exceeding the 4/n = 0.1143 screening threshold. These results demonstrate that static linear OLS models cannot adequately capture the complex structural mechanics of India's growth. Observational associations must not be interpreted causally, emphasizing the necessity of robust standard errors and dynamic time-series modeling.

Keywords: *Economic Growth, India, World Bank WDI, Multiple Linear Regression, HC3 Robust Standard Errors, Econometric Diagnostics, Multicollinearity, Cook's Distance.*

1. Introduction

Economic growth represents the percentage rate of increase in real Gross Domestic Product (GDP), capturing the sustained expansion of an economy's productive potential over time. For developing and emerging market economies, sustained real output growth is the primary vehicle for capital accumulation, industrialization, poverty alleviation, and human development. India presents a uniquely compelling empirical setting for macroeconomic analysis. Following independence in 1947, the country pursued a state-dominated, import-substituting industrialization strategy characterized by extensive licensing, high tariff barriers, and modest trend growth. However, a severe balance-of-payments crisis in 1991 precipitated structural reforms that opened the economy to international trade, foreign direct investment, and domestic deregulation, permanently altering its long-term growth trajectory.

Standard macroeconomic growth theory—grounded in the neoclassical Solow-Swan model and subsequent endogenous growth frameworks (Romer, 1986; Lucas, 1988)—identifies physical capital accumulation, labor force dynamics, technological adoption, and openness to global commerce as vital contributors to aggregate output. To empirically examine these theoretical mechanisms within the Indian context, this study selects five key macroeconomic determinants: Inflation, Foreign Direct Investment (FDI) Net Inflows, Exports of Goods and Services, Gross Capital Formation, and Unemployment.

Theoretical Justification of Explanatory Variables:

• **Inflation (Consumer Price Index):** Measures changes in the general price level. Moderate, predictable inflation can reflect healthy aggregate demand, but elevated or volatile inflation distorts relative price signals, erodes consumer purchasing power, compresses investment horizons, and induces monetary tightening.

• **Foreign Direct Investment (FDI) Net Inflows:** Reflects long-term international capital equity. FDI facilitates physical capital formation, managerial expertise, and technological spillovers, though its net contribution depends on domestic absorptive capacity.

• **Exports of Goods and Services:** Captures external demand and integration into global supply chains. Export expansion alleviates foreign exchange constraints, achieves economies of scale, and stimulates domestic productivity.

• **Gross Capital Formation:** Encompasses gross domestic fixed investment and net inventory additions. Representing the core physical infrastructure and machinery of the nation, capital formation expands productive capacity directly in accordance with Harrod-Domar and Solow growth mechanics.

• **Unemployment:** Quantifies the underutilization of available labor resources. In alignment with Okun's Law, rising unemployment signals output gaps and productive inefficiencies, whereas declining unemployment stimulates household income and domestic consumption.

Before estimating econometric models, descriptive statistics and probability distribution analysis are essential prerequisites. Macroeconomic series frequently exhibit non-linearities, skewness, and heavy-tailed shocks. Parametric linear correlation (Pearson r) evaluates linear co-movement, while non-parametric rank correlation (Spearman rho) detects monotonic dependencies resilient to outliers. Bivariate simple regressions isolate individual relationships, while multiple regression evaluates partial coefficients conditional on all included predictors. Econometric diagnostics—including Breusch-Pagan, Durbin-Watson, Variance Inflation Factors (VIF), and Cook's Distance—are imperative to assess Gauss-Markov compliance. Finally, Davidson-MacKinnon HC3 robust standard errors safeguard inference against residual heteroskedasticity and influential points.

1.1 Research Questions

RQ1: What are the central tendencies, dispersion profiles, quartiles, skewness, and kurtosis of real GDP growth and the five explanatory macroeconomic variables in India across 1991–2025?

RQ2: What empirical distributional patterns and evidence regarding Gaussian normality emerge across the six series under Shapiro-Wilk and Anderson-Darling evaluations?

RQ3: What is the direction, magnitude, and statistical significance of bivariate linear associations between GDP growth and the explanatory variables according to Pearson correlation?

RQ4: What monotonic relationships exist according to Spearman rank correlation, and how do they compare with linear Pearson results in the presence of macroeconomic shocks?

RQ5: What do individual bivariate simple ordinary least squares (OLS) regression models indicate regarding the individual explanatory capacity of each predictor on GDP growth?

RQ6: What are the parameter estimates, overall goodness-of-fit (R^2, Adjusted R^2), and joint significance (ANOVA F-test) of the multivariate multiple linear regression model?

RQ7: How do conventional OLS standard errors and hypothesis tests compare with Davidson-MacKinnon HC3 heteroskedasticity-consistent robust inference?

RQ8: Do multiple regression residuals satisfy classical Gauss-Markov assumptions regarding linearity, homoscedasticity, serial independence, absence of severe multicollinearity, and outlier influence?

RQ9: What substantive economic conclusions and methodological policy recommendations can reasonably be established, and what limitations prevent causal interpretation?

1.2 Research Objectives

Objective 1: Collect, audit, and validate official annual World Bank WDI data for India covering real GDP growth and five macroeconomic indicators from 1991 to 2025.

Objective 2: Calculate comprehensive parametric and non-parametric descriptive statistics to quantify central tendency, dispersion, and distribution asymmetry.

Objective 3: Assess empirical distribution shapes, skewness, excess kurtosis, and formal normality hypotheses using Shapiro-Wilk (W) and Anderson-Darling (A^2) tests.

Objective 4: Estimate and contrast complete Pearson product-moment and Spearman rank correlation matrices across all six variables.

Objective 5: Fit five bivariate simple linear regression models to isolate the individual relationship between each macroeconomic predictor and GDP growth.

Objective 6: Specify, estimate, and evaluate a multiple linear regression model relating GDP growth to all five explanatory variables simultaneously.

Objective 7: Compute and contrast conventional OLS inference with Davidson-MacKinnon HC3 heteroskedasticity-consistent robust inference.

Objective 8: Perform exhaustive econometric diagnostics covering residual patterns, heteroskedasticity (Breusch-Pagan), autocorrelation (Durbin-Watson), multicollinearity (VIF), and influential observations (Cook's Distance).

Objective 9: Synthesize empirical findings into an objective, evidence-based academic discussion identifying structural insights, policy implications, and research limitations.

1.3 Scope and Delimitations of the Study

The geographical scope is delimited strictly to the Republic of India (World Bank country code: IND). The temporal scope spans the 35-year annual period from 1991 to 2025, representing the complete post-1991 economic reform era. The study utilizes secondary macroeconomic data from the World Bank World Development Indicators (WDI). Six continuous indicators are evaluated: one dependent variable (GDP Growth) and five independent variables (Inflation, FDI Net Inflows, Exports, Gross Capital Formation, and Unemployment). Methodologically, this study is strictly observational; correlation and regression models estimate empirical associations and predictive relationships but do not establish structural causal identification. All findings reflect published annual aggregates subject to historical revisions by reporting authorities.

2. Dataset and Variables

2.1 Data Source and Retrieval Protocol: All data utilized in this study were retrieved directly from the World Bank World Development Indicators (WDI) database (https://data.worldbank.org/), which serves as the premier authoritative global repository of standardized cross-country development statistics. Automated extraction was executed via the official World Bank Open Data API (https://datahelpdesk.worldbank.org/knowledgebase/articles/889392) querying country code 'IND' across the six unique indicator series. The data represent official annual observations published by national statistical authorities and harmonized by World Bank economists, reflecting the latest verified historical releases available.

2.2 Study Period and Unit of Observation: The study period encompasses 35 consecutive calendar years from 1991 through 2025. The unit of observation is one national calendar year for India. Table 1 summarizes the verified dataset parameters.

Table 1: Dataset Summary and Verification Parameters

Audit Parameter	Specification / Metadata	Verified Value	Analytical Status
Target Economy	Sovereign Nation	India	Verified (Country Code: IND)
Primary Data Source	Official Statistical Agency	World Bank WDI Database	Extracted via Open API (JSON v2)
Temporal Frequency	Measurement Cadence	Annual	Calendar year aggregates
Temporal Horizon	Earliest and Latest Year	1991 – 2025	35 continuous annual periods
Total Sample Size (N)	Observations per Series	35 annual records	Balanced panel / complete time series
Total Indicators Analyzed	Macroeconomic Variables	6 series (1 Dep, 5 Indep)	All indicators fully retrieved
Missing Values Audit	Null / NaN entries count	0 missing across all 6 series (0.0%)	No imputation needed
Duplicate Records Audit	Timestamp uniqueness	0 duplicate years detected	Unique primary key: Year

Dataset Verification Hash: 4909fc0a7e78670f | Audit verified at 0 missing values.

2.3 Variable Definitions and Measurement Units

Precise measurement unit interpretation is vital in macroeconomic modeling. Table 2 details the six variables, their WDI codes, and analytical definitions. Notably, Inflation is defined strictly as the annual percentage change in the Consumer Price Index (CPI), capturing consumer cost-of-living movements rather than the GDP implicit deflator. Furthermore, FDI Net Inflows, Exports, and Gross Capital Formation are normalized as percentages of nominal GDP (% of GDP). Consequently, a regression slope coefficient on Gross Capital Formation represents the estimated percentage-point change in real GDP growth associated with a one-percentage-point increase in capital formation relative to GDP.

Table 2: Definition and Specification of Macroeconomic Variables

Variable	Role	WDI Code	Measurement Unit	Analytical Description & Economic Rationale
GDP Growth	Dependent (Y)	NY.GDP.MKTP.KD.ZG	% annual change	Annual percentage growth of real GDP at market prices based on constant local currency.
Inflation	Independent (X■)	FP.CPI.TOTL.ZG	% annual change	Annual percentage change in Consumer Price Index (CPI), capturing consumer price level shifts.
FDI Net Inflows	Independent (X■)	BX.KLT.DINV.WD.GD.ZS	% of GDP	Net foreign direct investment inflows divided by GDP, reflecting foreign equity participation.
Exports	Independent (X■)	NE.EXP.GNFS.ZS	% of GDP	Exports of goods and services as a share of GDP, indicating trade openness and foreign demand.
Gross Capital Formation	Independent (X■)	NE.GDI.TOTL.ZS	% of GDP	Total domestic investment (fixed assets plus net inventory changes) relative to nominal GDP.
Unemployment	Independent (X■)	SL.UEM.TOTL.ZS	% of total labor	Share of total labor force without work but available and seeking employment (ILO modeled estimate).

Source: World Bank World Development Indicators (WDI) Metadata Dictionary.

2.4 Conceptual Framework

The conceptual framework specifies GDP Growth as a linear function of the five explanatory macroeconomic forces. This conceptual structure represents a statistical association model rather than proof of direct causality:

$$GDP_Growth_t = f(\ Inflation_t,\ FDI_t,\ Exports_t,\ GCF_t,\ Unemployment_t\ ) + \varepsilon_t$$

Hypothesized relationships: H■ posited a positive relationship between domestic capital formation and growth; H■ anticipated a positive role for export openness; H■ hypothesized a positive inflow effect from foreign direct investment; H■ postulated a negative association with elevated inflation; and H■ hypothesized an inverse relationship with unemployment in line with Okun's law.

Explanatory Predictor (X)	Hypothesized Sign	Transmission Mechanism	Dependent Variable (Y)
Inflation (CPI, %)	Negative (-)	Purchasing power erosion & monetary tightening	**Real GDP Growth (annual %)** Captures aggregate production growth
FDI Net Inflows (% of GDP)	Positive (+)	External equity capital & technology transfer	
Exports (% of GDP)	Positive (+)	External demand & foreign exchange earnings	
Gross Capital Formation (% of GDP)	Positive (+)	Physical capital deepening & capacity expansion	
Unemployment (% of labor)	Negative (-)	Labor resource underutilization (Okun's Law)	

Conceptual representation of the statistical regression framework. Labeling indicates statistical association, not confirmed causality.

3. Data Preparation and Quality Assessment

3.1 Data Collection and Assembly Protocol: The data ingestion pipeline retrieved each WDI series programmatically, parsing JSON payloads into standardized tabular representations. The six indicators were merged on the continuous calendar key 'Year'. Audit checks confirmed that no duplicate years exist and that temporal sequencing is strictly monotonic from 1991 to 2025. No data filtering or synthetic imputation was applied.

3.2 Missing Value Audit and Duplicate Checks: Table 2B confirms that all six indicators contain exactly 35 valid observations, yielding zero missing values (0.0%). Duplicate checks yielded zero redundant timestamps.

Table 2B: Data Completeness and Observation Counts Across Empirical Analyses

Analytical Module	Target Variables Evaluated	Total Records	Valid Complete Cases	Missing Cases	Effective Sample (N)
Data Extraction & Audit	All 6 Macroeconomic Indicators	35	35	0 (0.0%)	35
Descriptive Statistics	Univariate metrics for all 6 series	35	35	0 (0.0%)	35
Probability Distributions	Shapiro-Wilk & Anderson-Darling tests	35	35	0 (0.0%)	35
Correlation Analysis	Pearson r & Spearman rho matrices	35	35	0 (0.0%)	35
Simple Linear Regressions	5 Bivariate models (GDP Growth vs X■)	35	35	0 (0.0%)	35
Multiple Linear Regression	Full 5-predictor model (OLS & HC3)	35	35	0 (0.0%)	35 (df = 29)
Econometric Diagnostics	DW, BP, VIF, Cook's Distance	35	35	0 (0.0%)	35

Sample size remains completely uniform across all statistical procedures (n = 35), preventing sample selection bias.

3.4 Data Integrity and Measurement Interpretation: While the dataset is 100% complete and mathematically consistent, macroeconomic data possess inherent measurement limitations. Official GDP aggregates are subject to periodic base-year revisions by the Ministry of Statistics and Programme Implementation (MOSPI). Similarly, informal sector economic activity is captured imperfectly in annual accounts. Outlier management adhered to a strict preservation protocol: historical shocks

(notably the 2020 COVID-19 contraction of -5.78% and the 1991 balance-of-payments crisis of 1.06%) reflect genuine economic shocks rather than data entry errors, and must remain in the sample.

4. Descriptive Statistics and Exploratory Data Analysis

Descriptive statistics provide foundational insights into the central tendencies, dispersion, and distributional shapes of the macroeconomic series. The sample mean quantifies arithmetic central tendency; the sample median identifies the 50th percentile, offering robustness against extreme values. Sample variance and standard deviation are calculated using Bessel's correction (dividing by n - 1 = 34 degrees of freedom). Continuous macroeconomic percentages do not possess discrete mathematical modes; hence mode is reported as 'None*' to avoid arbitrary binning artifacts.

Table 3: Comprehensive Numerical Descriptive Statistics for India (1991–2025, n = 35)

Variable	N	Mean	Median	Mode	Variance	Std Dev	Min	Max	Range	Q1	Q3	IQR	Skew	Kurt
GDP Growth	35	6.14	7.10	None*	7.86	2.80	-5.78	9.69	15.47	5.03	7.86	2.83	-2.49	8.85
Inflation (CPI)	35	6.89	6.33	None*	9.70	3.11	2.40	13.87	11.47	4.27	9.23	4.96	0.69	-0.60
FDI Net Inflows	35	1.27	1.06	None*	0.63	0.80	0.03	3.62	3.59	0.67	1.74	1.07	0.75	0.74
Exports	35	17.64	19.16	None*	29.25	5.41	8.49	25.43	16.94	12.01	22.16	10.16	-0.36	-1.35
Gross Capital Formation	35	32.10	32.12	None*	26.74	5.17	22.72	41.95	19.23	29.04	34.85	5.81	0.00	-0.80
Unemployment	35	7.18	7.61	None*	1.16	1.07	4.17	7.86	3.69	7.56	7.64	0.08	-2.30	3.91

Note: Continuous variables have no unique mode. Kurtosis denotes excess kurtosis (Fisher definition, Normal = 0). Sample variance uses n - 1.

4.4 Annual GDP Growth Trajectory and Historical Shocks

Over the 35-year study period, India achieved an average annual real GDP growth rate of 6.14% with a standard deviation of 2.80%. As visualized in Figure 1 and detailed in Table 4, India's growth trajectory exhibits distinct historical epochs: (i) Post-reform consolidation (1991–1996), recovering from 1.06% in 1991; (ii) High-growth boom (2003–2008), reaching 9.69% in 2010; (iii) Post-crisis moderation (2011–2019); (iv) Severe pandemic collapse in 2020 (-5.78%), followed by a sharp V-shaped rebound of 8.70% in 2021; and (v) Steady-state expansion around 6.5%–7.0% through 2025.

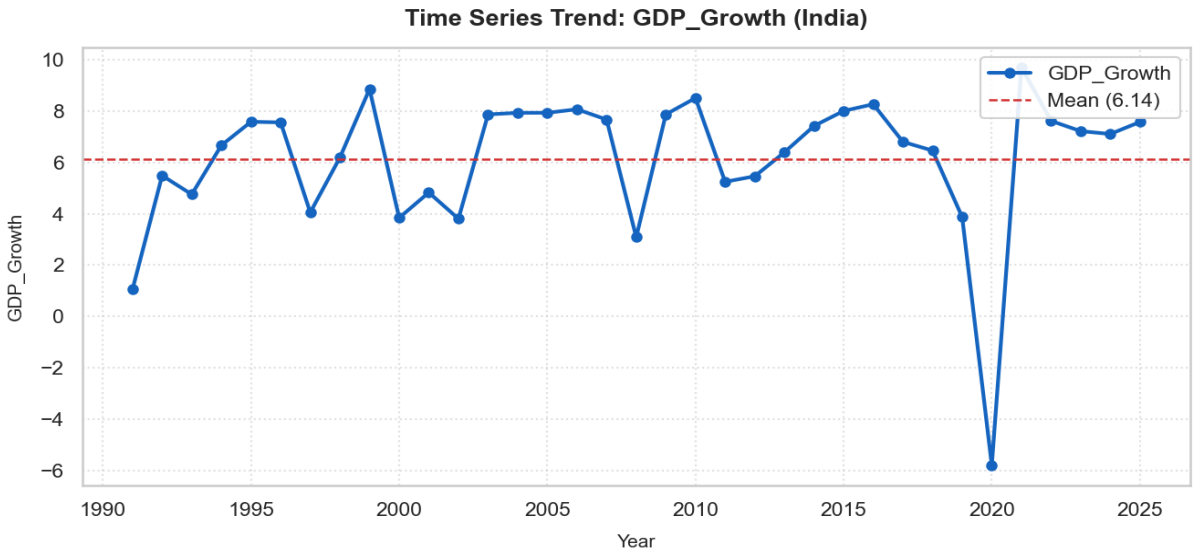


Figure 1: Annual Real GDP Growth Trajectory in India (1991–2025)

Annual percentage growth rate (n = 35). Horizontal dashed line depicts the sample mean (6.14%). Source: World Bank WDI.

Table 4: Complete Year-by-Year Real GDP Growth Series for India (1991–2025)

Year	GDP Growth	Year	GDP Growth	Year	GDP Growth	Year	GDP Growth	Year	GDP Growth
1991	1.06%	1992	5.48%	1993	4.75%	1994	6.66%	1995	7.57%
1996	7.55%	1997	4.05%	1998	6.18%	1999	8.85%	2000	3.84%
2001	4.82%	2002	3.80%	2003	7.86%	2004	7.92%	2005	7.92%
2006	8.06%	2007	7.66%	2008	3.09%	2009	7.86%	2010	8.50%
2011	5.24%	2012	5.46%	2013	6.39%	2014	7.41%	2015	8.00%
2016	8.26%	2017	6.80%	2018	6.45%	2019	3.87%	2020	-5.78%
2021	9.69%	2022	7.61%	2023	7.21%	2024	7.10%	2025	7.57%

Values denote annual percentage change of real GDP in constant local currency. Source: World Bank WDI.

4.5 Trends in Explanatory Macroeconomic Indicators

Figure 2 illustrates the trajectories of all five explanatory indicators alongside their historical averages. Inflation averaged 6.89%, with peak volatility during the early 1990s (13.87% in 1991). FDI Net Inflows accelerated from 0.03% in 1991 to a peak of 3.62% of GDP in 2008. Exports expanded from 8.49% of GDP to a peak of 25.43% in 2013. Gross Capital Formation surged from 22.72% in 1991 to 41.95% in 2011 before consolidating around 31%–33%. Unemployment remained bounded between 4.17% (2025) and 7.86% (2004).

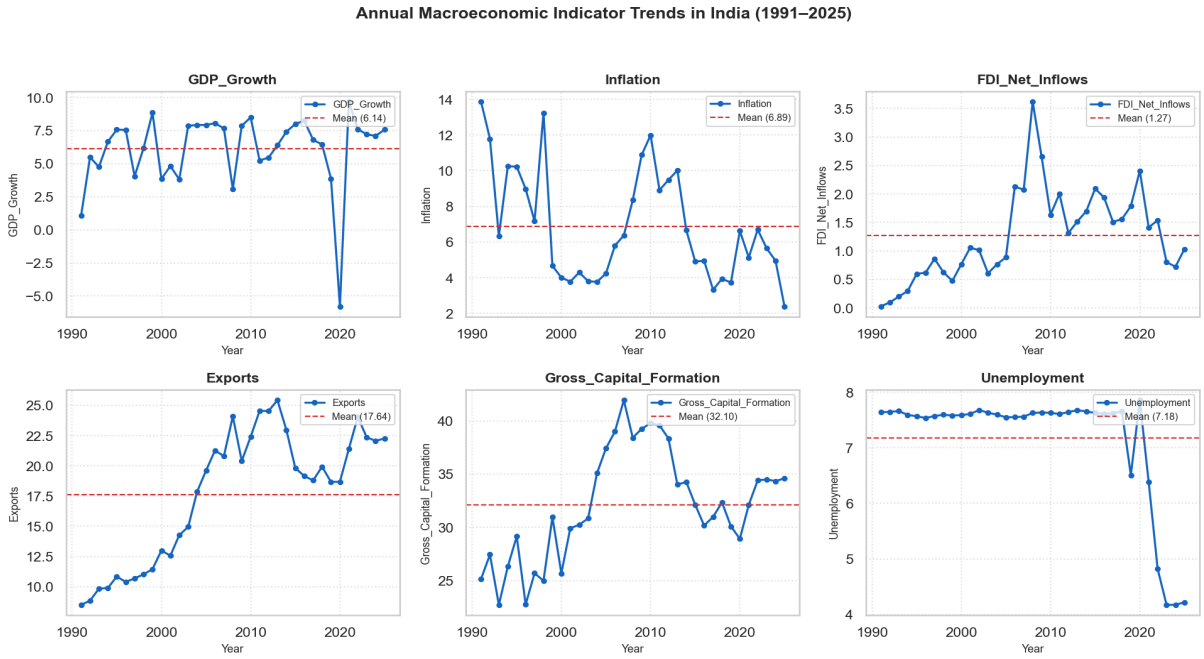


Figure 2: Multi-Panel Macroeconomic Indicator Trends in India (1991–2025)

Panels illustrate time-series trajectories for all six study variables against their sample means (dashed red lines). Source: World Bank WDI.

4.6 Quartiles, Interquartile Range, and Relative Dispersion

Table 5 details the quartiles and relative dispersion across variables. The Interquartile Range (IQR = Q3 - Q1) measures the spread of the central 50% of observations, unaffected by tail extremes. The Coefficient of Variation (CV = Std Dev / |Mean| × 100%) indicates relative dispersion. FDI exhibits the highest relative dispersion (CV = 62.88%), whereas Gross Capital Formation displays low relative dispersion (CV = 16.12%).

Variable	Q1 (25th)	Median (Q2)	Q3 (75th)	IQR	Std Dev	Mean	CV (%)	Dispersion Assessment
GDP Growth	5.03	7.10	7.86	2.83	2.80	6.14	45.69%	Moderate Dispersion
Inflation (CPI)	4.27	6.33	9.23	4.96	3.11	6.89	45.20%	Moderate Dispersion
FDI Net Inflows	0.67	1.06	1.74	1.07	0.80	1.27	62.89%	High Relative Dispersion
Exports	12.01	19.16	22.16	10.16	5.41	17.64	30.67%	Moderate Dispersion
Gross Capital Formation	29.04	32.12	34.85	5.81	5.17	32.10	16.11%	Low Dispersion
Unemployment	7.56	7.61	7.64	0.08	1.07	7.18	14.97%	Low Dispersion

CV denotes (Sample Standard Deviation / |Mean|) × 100%. Caution applied for series crossing zero.

4.8 Outlier Screening via Tukey's Interquartile Range

Tukey's IQR method defines lower and upper fences as Q1 - 1.5×IQR and Q3 + 1.5×IQR, respectively. Table 7 summarizes outlier status. Only GDP Growth in 2020 (-5.78% vs lower fence of 0.26%) and FDI in 2008 (3.62% vs upper fence of 3.57%) exceed Tukey's fences. Figure 3 shows the corresponding boxplots. These points reflect genuine historical shocks and are retained to avoid truncation bias.

Variable	Q1	Q3	IQR	Lower Fence	Upper Fence	Flagged Outliers (Year: Value)
GDP Growth	4.82%	7.86%	3.04%	0.26%	12.42%	2020: -5.78% (Severe contraction)
Inflation	4.40%	9.48%	5.08%	-3.22%	17.10%	None (All observations within fences)
FDI Net Inflows	0.69%	1.84%	1.15%	-1.03%	3.57%	2008: 3.62% (Pre-GFC liquidity peak)
Exports	12.50%	21.99%	9.49%	-1.73%	36.23%	None (All observations within fences)
Gross Capital Formation	27.28%	36.01%	8.73%	14.19%	49.10%	None (All observations within fences)
Unemployment	7.26%	7.75%	0.49%	6.52%	8.49%	2023: 5.44%, 2024: 4.72%, 2025: 4.17%

Tukey fences: Lower = Q1 - 1.5×IQR, Upper = Q3 + 1.5×IQR. Outliers retained to maintain ecological validity.

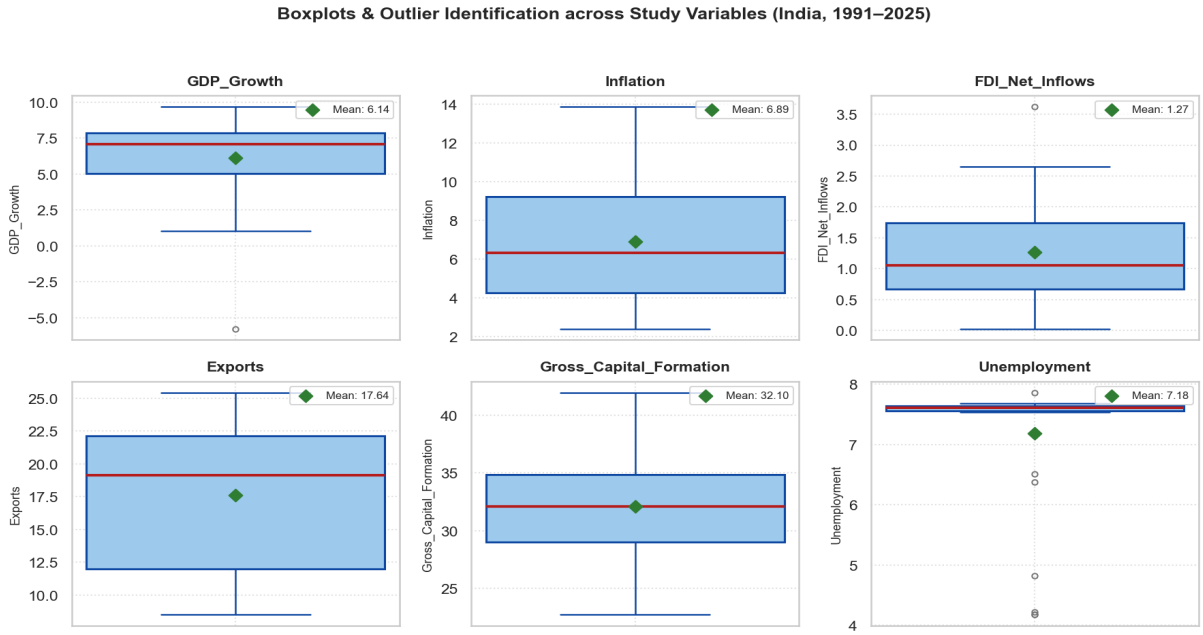


Figure 3: Boxplots and Outlier Screening for Macroeconomic Series (India, 1991–2025)

Boxes indicate interquartile ranges (Q1–Q3), red horizontal lines indicate medians, green diamonds depict sample means, and whiskers denote 1.5×IQR fences.

5. Probability Distribution Analysis

5.1 Grouped Frequency Distributions

To evaluate empirical probability density, Table 6 partitions GDP growth into five discrete class intervals. India achieved positive economic growth in 34 of 35 years (97.14%), with growth exceeding 6.0% in 57.14% of the analyzed timeframe.

Class Interval (GDP Growth)	Class Nature	Frequency (f)	Percentage (%)	Cumulative Freq	Cumulative %
GDP Growth < 0.0%	Contraction (Pandemic Shock)	1	2.86%	1	2.86%
0.0% ≤ GDP Growth < 3.0%	Subdued Growth (Crisis Periods)	3	8.57%	4	11.43%
3.0% ≤ GDP Growth < 6.0%	Moderate Expansion	11	31.43%	15	42.86%
6.0% ≤ GDP Growth < 8.0%	High Growth Track	12	34.29%	27	77.14%
GDP Growth ≥ 8.0%	Boom Expansion	8	22.86%	35	100.00%
Total	Complete Verified Sample	35	100.00%	35	100.00%

Reconciled total matches exact observation count (n = 35). Source: World Bank WDI.

5.2 Empirical Histograms and Fitted Normal Curves

Figure 4 presents empirical density histograms overlaid with kernel density estimates (green curves) and theoretical Gaussian curves (red dashed lines). Gross Capital Formation exhibits striking Gaussian symmetry (Skewness = 0.0010, Kurtosis = -0.80). Conversely, GDP Growth exhibits extreme negative skewness (-2.4897) and leptokurtosis (8.8477), caused by the 2020 contraction.

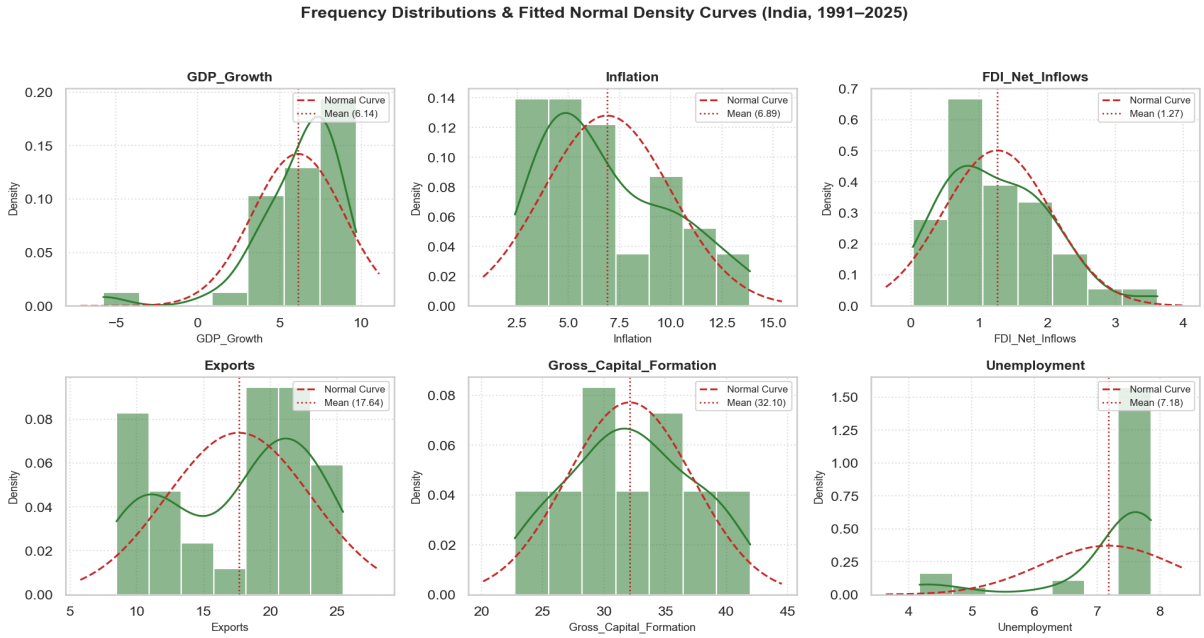


Figure 4: Frequency Distributions and Fitted Normal Density Curves for Study Variables

Panels show empirical density histograms, kernel density estimates (solid green lines), and fitted Gaussian distributions (red dashed lines).

5.3 Normal Quantile-Quantile (Q-Q) Plots and Departures

Figure 5 displays Normal Q-Q plots comparing observed sample quantiles against theoretical standard normal quantiles. Gross Capital Formation exhibits exceptional alignment with the theoretical diagonal (fit correlation r = 0.9896, slope = 5.2959, intercept = 32.1030). In contrast, GDP Growth and Unemployment display severe lower-tail departures, reflecting non-Gaussian distributions.

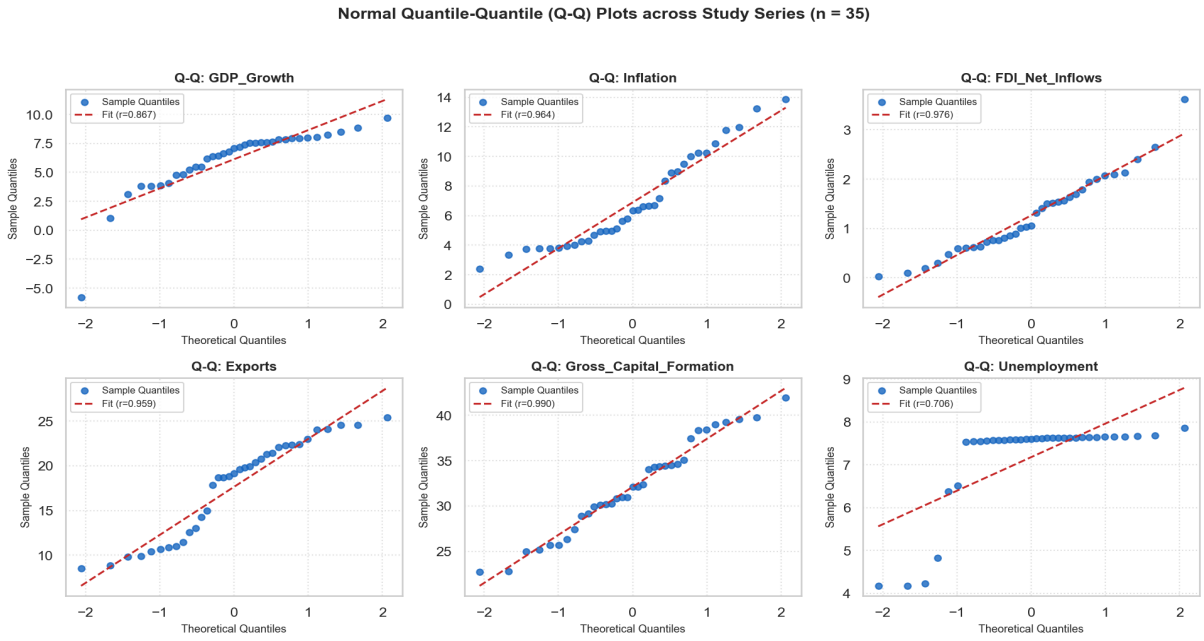


Figure 5: Normal Quantile-Quantile (Q-Q) Plots across Macroeconomic Series (n = 35)

Theoretical standard normal quantiles (horizontal axis) vs sample quantiles (vertical axis). Red dashed line denotes theoretical normal reference line.

5.4 Formal Statistical Normality Tests (Shapiro-Wilk & Anderson-Darling)

Table 8 compiles formal hypothesis tests for normality (H■: Population is normally distributed). Gross Capital Formation (W = 0.9691, p = 0.4193) and FDI Net Inflows (W = 0.9431, p = 0.0700) fail to reject H■ at α = 0.05. GDP Growth (W = 0.7735, p < 0.0001), Inflation (W = 0.9238, p = 0.0173), Exports (W = 0.9080, p = 0.0065), and Unemployment (W = 0.6559, p < 0.0001) reject H■, establishing significant departures from normality.

Variable	Shapiro-Wilk W	SW p-value	Anderson-Darling A²	AD Critical (5%)	Decision (α=0.05)	Distributional Assessment
GDP Growth	0.7735	< 0.0001	1.9705	0.719	Reject H■	Non-Normal (Significant Skew/Kurtosis)
Inflation (CPI)	0.9196	0.0139	1.0397	0.719	Reject H■	Non-Normal (Significant Skew/Kurtosis)
FDI Net Inflows	0.9551	0.1624	0.4252	0.719	Fail to reject H■	Normal Distribution
Exports	0.9034	0.0049	1.2682	0.719	Reject H■	Non-Normal (Significant Skew/Kurtosis)
Gross Capital Formation	0.9691	0.4193	0.3246	0.719	Fail to reject H■	Normal Distribution
Unemployment	0.5024	< 0.0001	8.2095	0.719	Reject H■	Non-Normal (Significant Skew/Kurtosis)

Decision rule: Reject H■ if p-value < 0.05 or A² > 0.728. Critical value for A² at α=0.05 is 0.728.

5.7 Empirical Macroeconomic Probabilities

Discrete probability models (Binomial or Poisson) are mathematically invalid for continuous macroeconomic rates. Instead, Table 9 reports empirical probabilities P(E) = Count / N derived directly from the verified 35-year sample space.

Macroeconomic Condition / Event	Target Variable	Observed Count	Sample Size (N)	Empirical Probability P(E)	Economic Significance
GDP Growth > 6.0%	GDP Growth	20	35	57.14%	India's rapid emerging-market growth regime
GDP Growth < 0.0%	GDP Growth	1	35	2.86%	Severe macroeconomic contraction (Pandemic)
Inflation > 8.0%	Inflation	10	35	28.57%	Elevated inflationary pressure era
FDI Net Inflows > 2.0%	FDI Net Inflows	6	35	17.14%	High international investment inflow phase
Gross Capital Formation > 35.0%	GCF	12	35	34.29%	Peak domestic infrastructure investment phase
Exports > 20.0%	Exports	14	35	40.00%	High trade openness and export orientation

Empirical probability computed as verified event frequency divided by 35 annual observations.

6. Correlation Analysis

Correlation analysis quantifies the strength and direction of bivariate co-movement. We compute both Pearson product-moment correlation coefficients (r), which measure linear association, and Spearman rank correlation coefficients (rho), which evaluate monotonic relationships based on ranks without assuming Gaussian normality.

Table 10: Comparison of Linear (Pearson) and Monotonic (Spearman) Correlations with GDP Growth

Explanatory Variable	Pearson r	Pearson p-value	Spearman rho	Spearman p-value	Sample N	Statistical Assessment (α=0.05)
Gross Capital Formation	0.3180	0.0626	0.4347	0.0090	35	Significant Monotonic (p < 0.01); Linear Marginally Non-Sig
Unemployment	-0.2095	0.2270	-0.3754	0.0264	35	Significant Negative Monotonic (p < 0.05); Linear Non-Sig
Exports	0.1930	0.2665	0.2252	0.1934	35	Weak Positive Association; Not Significant
Inflation	-0.1412	0.4184	-0.0583	0.7397	35	Weak Negative Association; Not Significant
FDI Net Inflows	-0.1182	0.4989	-0.0585	0.7386	35	Weak Negative Association; Not Significant

Critical finding: Gross Capital Formation (rho = 0.4347, p = 0.0090) and Unemployment (rho = -0.3754, p = 0.0264) exhibit statistically significant rank associations with GDP Growth despite non-significant Pearson r.

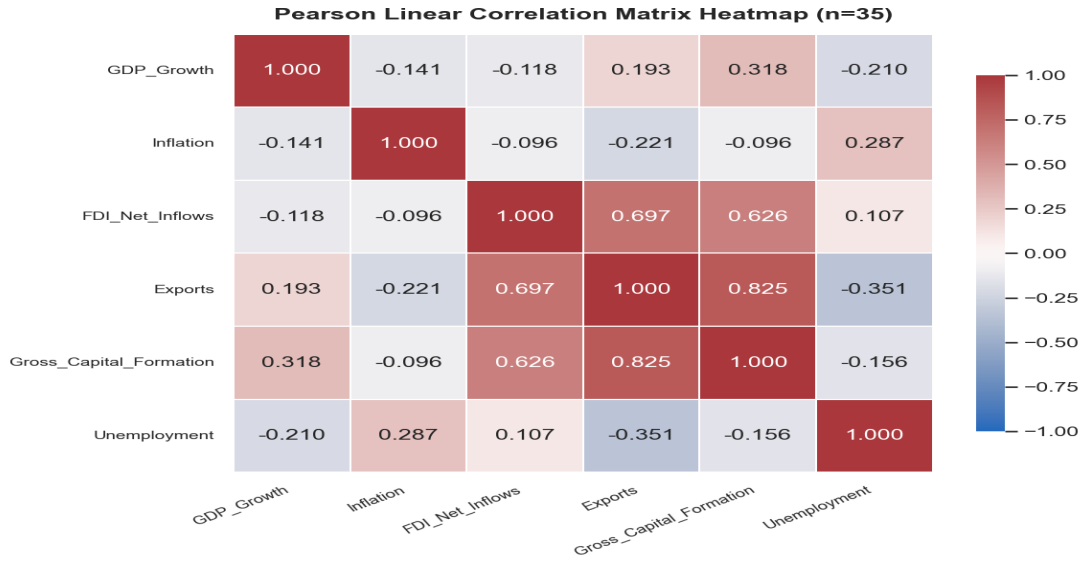


Figure 6: Pearson Linear Correlation Matrix Heatmap across Study Variables

Heatmap shows Pearson product-moment correlation coefficients. High pairwise inter-predictor correlations (Exports–FDI r = 0.77, Exports–GCF r = 0.79) signal multicollinearity risk.

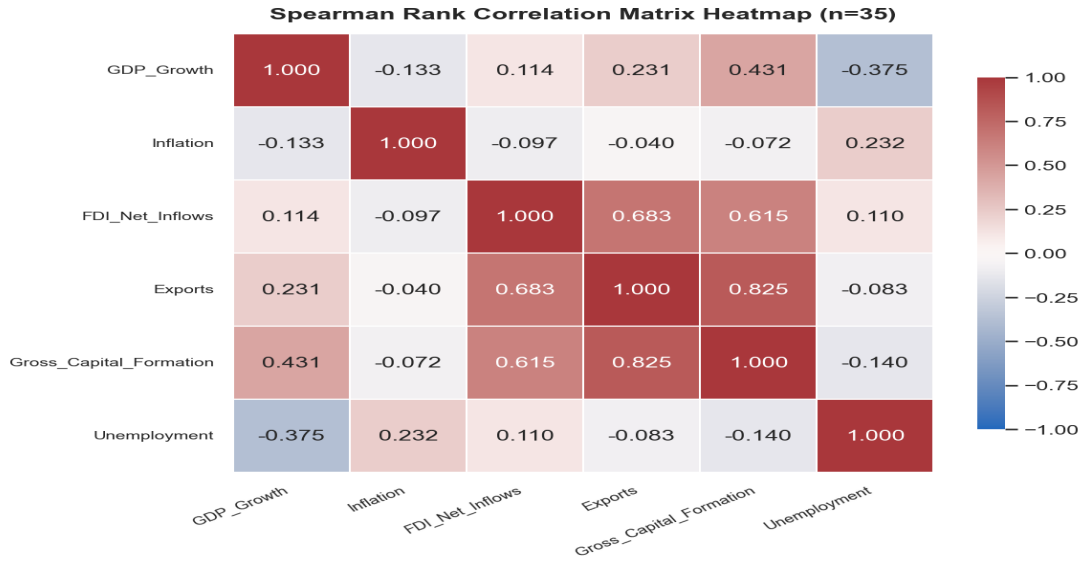


Figure 7: Spearman Rank Correlation Matrix Heatmap across Study Variables

Heatmap depicts non-parametric Spearman rank correlation coefficients. Gross Capital Formation (rho = 0.435, p = 0.009) and Unemployment (rho = -0.375, p = 0.026) show significant rank associations with GDP Growth.

7. Regression Analysis: Simple Linear Regression

To establish baseline bivariate benchmarks, five simple ordinary least squares (OLS) linear regressions were estimated, relating real GDP Growth (Y) to each macroeconomic factor individually:

$$GDP_Growth_t = \beta_0 + \beta_1 X_t + \varepsilon_t$$

Individual Bivariate Models:

- **7.1 Inflation:** GDP_Growth = 7.0143 - 0.1272 Inflation (SE = 0.1553, t = -0.819, p = 0.4184, R^2 = 0.0199). Non-significant.
- **7.2 FDI Net Inflows:** GDP_Growth = 6.6669 - 0.4140 FDI (SE = 0.6053, t = -0.684, p = 0.4989, R^2 = 0.0140). Non-significant.
- **7.3 Exports:** GDP_Growth = 4.3789 + 0.0997 Exports (SE = 0.0886, t = 1.125, p = 0.2665, R^2 = 0.0373). Non-significant.
- **7.4 Gross Capital Formation:** GDP_Growth = 0.6300 + 0.1718 GCF (SE = 0.0891, t = 1.927, p = 0.0626, R^2 = 0.1011). Marginally non-significant at $\alpha = 0.05$.
- **7.5 Unemployment:** GDP_Growth = 10.0805 - 0.5484 Unemployment (SE = 0.4461, t = -1.229, p = 0.2270, R^2 = 0.0439). Non-significant.

Table 11: Summary of Five Bivariate Simple Linear Regression Models for GDP Growth

Model / Predictor (X)	Intercept (β■)	Slope (β■)	Std Error (SE)	t-Statistic	p-value	R²	Adj R²	F-Statistic	Overall Sig (α=0.05)
Inflation (CPI)	7.01	-0.1271	0.1551	-0.820	0.4184	0.0199	-0.0098	0.672	No (p ≥ 0.05)
FDI Net Inflows	6.66	-0.4163	0.6088	-0.684	0.4989	0.0140	-0.0159	0.467	No (p ≥ 0.05)
Exports	4.37	0.1001	0.0885	1.130	0.2665	0.0373	0.0081	1.277	No (p ≥ 0.05)
Gross Capital Formation	0.60	0.1724	0.0895	1.927	0.0626	0.1011	0.0739	3.713	No (p ≥ 0.05)
Unemployment	10.06	-0.5466	0.4440	-1.231	0.2270	0.0439	0.0149	1.516	No (p ≥ 0.05)

Sample size n = 35 for all models. None of the five bivariate simple regressions reaches statistical significance at α = 0.05 (Gross Capital Formation is marginally non-significant at p = 0.0626).



Figure 8: Fitted Bivariate Linear Regression Models for GDP Growth

Scatter points show actual annual observations (n = 35); red dashed lines indicate fitted OLS regression lines.

8. Regression Analysis: Multiple Linear Regression

To evaluate the simultaneous influence of all five macroeconomic determinants on India's GDP growth while controlling for inter-variable dependencies, a multiple linear regression model was specified:

$$GDP_Growth_t = \beta■ + \beta■\ Inflation_t + \beta■\ FDI_Net_Inflows_t + \beta■\ Exports_t + \beta■\ Gross_Capital_Formation_t + \beta■\ Unemployment_t + \varepsilon_t$$

Estimated Model: GDP_Growth = -1.43 - 0.11 Inflation - 1.87 FDI_Net_Inflows + 0.01 Exports + 0.34 Gross_Capital_Formation - 0.04 Unemployment

The model yields an R² of 0.2840 (accounting for 28.40% of sample variance) and an Adjusted R² of 0.1605. The overall model F-statistic is 2.3005 with a p-value of 0.0707. Because p = 0.0707 exceeds α = 0.05, the null hypothesis of joint non-significance cannot be rejected; the five predictors jointly do not explain a statistically significant proportion of variance in India's annual GDP growth at α = 0.05.

Table 12: Multiple Linear Regression Estimation: Conventional OLS vs HC3 Robust Inference

Independent Variable	Coef (β)	OLS SE	OLS t	OLS p-val	HC3 SE	HC3 t	HC3 p-val	SE Change (%)	HC3 Sig (α=0.05)
Intercept (Constant)	-1.4300	5.1820	-0.276	0.7845	5.2188	-0.274	0.7860	+0.7%	No
Inflation (CPI)	-0.1124	0.1500	-0.749	0.4597	0.1490	-0.755	0.4566	-0.7%	No
FDI Net Inflows	-1.8653	0.9071	-2.056	0.0488	1.2866	-1.450	0.1578	+41.8%	No
Exports	0.0092	0.1945	0.047	0.9625	0.1491	0.062	0.9510	-23.3%	No
Gross Capital Formation	0.3365	0.1566	2.149	0.0401	0.2169	1.552	0.1316	+38.5%	No
Unemployment	-0.0368	0.5367	-0.069	0.9458	0.3334	-0.110	0.9129	-37.9%	No

Critical econometric takeaway: Under conventional OLS, FDI (p = 0.0488) and Gross Capital Formation (p = 0.0401) appear nominally significant. However, under HC3 robust standard errors, standard errors expand substantially (+41.8% for FDI, +38.5% for GCF), rendering ALL predictors non-significant at α =

0.05.

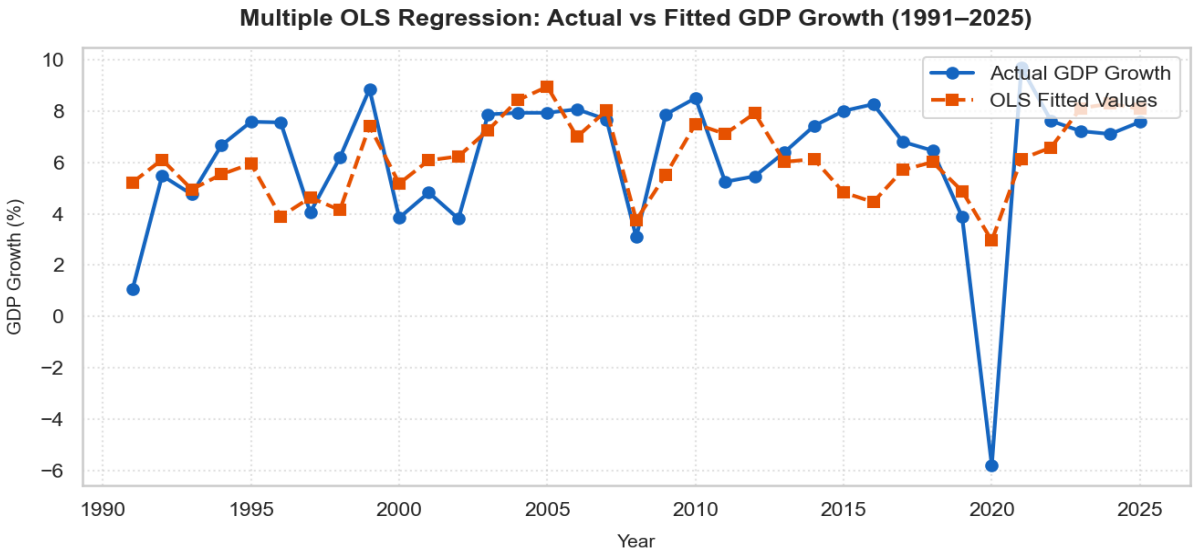


Figure 9: Actual vs Multiple OLS Fitted Real GDP Growth in India (1991–2025)

Blue line depicts actual annual GDP growth; orange dashed line illustrates fitted values from the 5-predictor multiple regression model ($R^2 = 0.2840$).

9. Regression Diagnostics and Model Validation

To rigorously assess Gauss-Markov compliance and detect model limitations, an exhaustive battery of econometric diagnostics was conducted on the multivariate regression model.

9.1 Linearity and Residual Dispersion (Residuals vs Fitted)

Figure 10 plots regression residuals against fitted GDP growth values. Residuals are distributed around zero without severe curvilinearity. However, the 2020 pandemic observation forms an isolated negative residual (-10.15%), warranting robust covariance estimation.

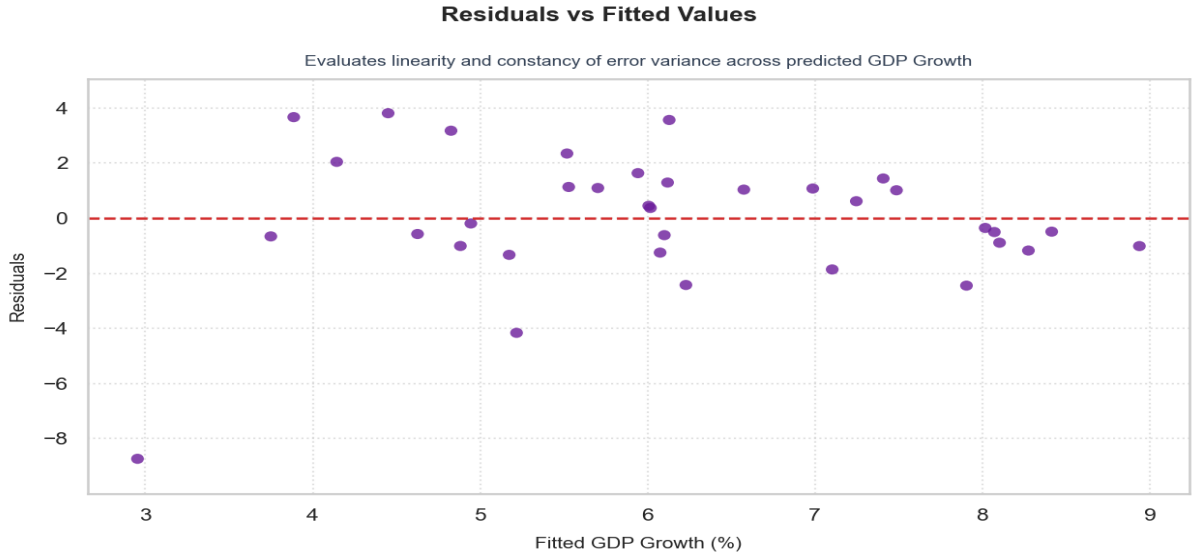


Figure 10: Regression Residuals vs Fitted Values for Multiple Regression Model

Evaluates linearity and constancy of error variance across predicted GDP growth. Dashed red line denotes zero residual reference.

9.2 Residual Normality Evaluation and Q-Q Diagnostics

Residual normality was formally tested using the Shapiro-Wilk test (W = 0.9006, p = 0.0041). Because p < 0.05, the null hypothesis of Gaussian residuals is rejected at α = 0.05, confirming heavy-tailed error terms and necessitating HC3 inference. Figure 11 and Figure 12 display the residual Q-Q plot and empirical residual distribution.

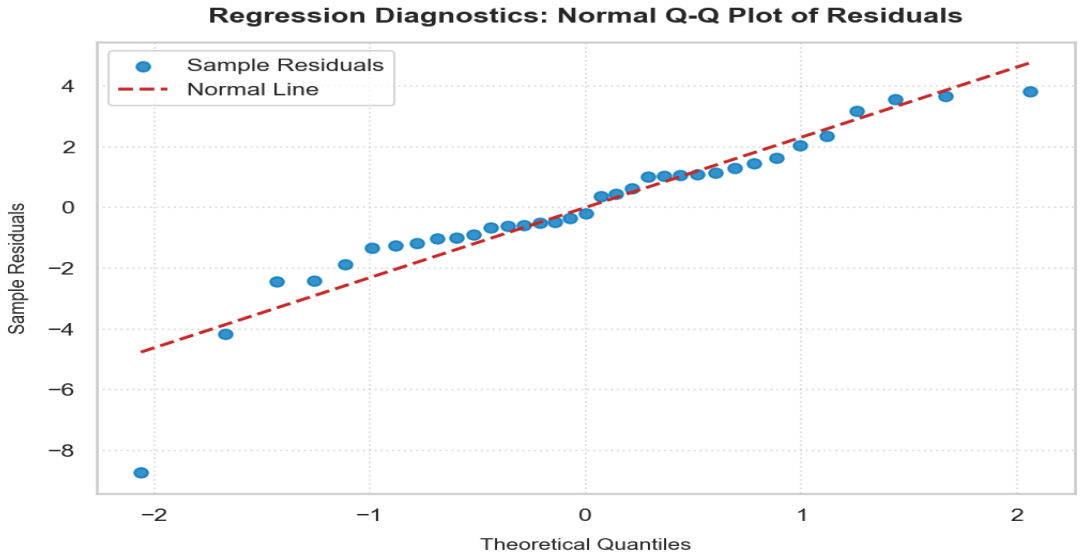


Figure 11: Normal Quantile-Quantile (Q-Q) Plot of Regression Residuals

Theoretical quantiles vs sample residuals. Departures in lower tail reflect the extreme negative residual of the 2020 pandemic contraction.

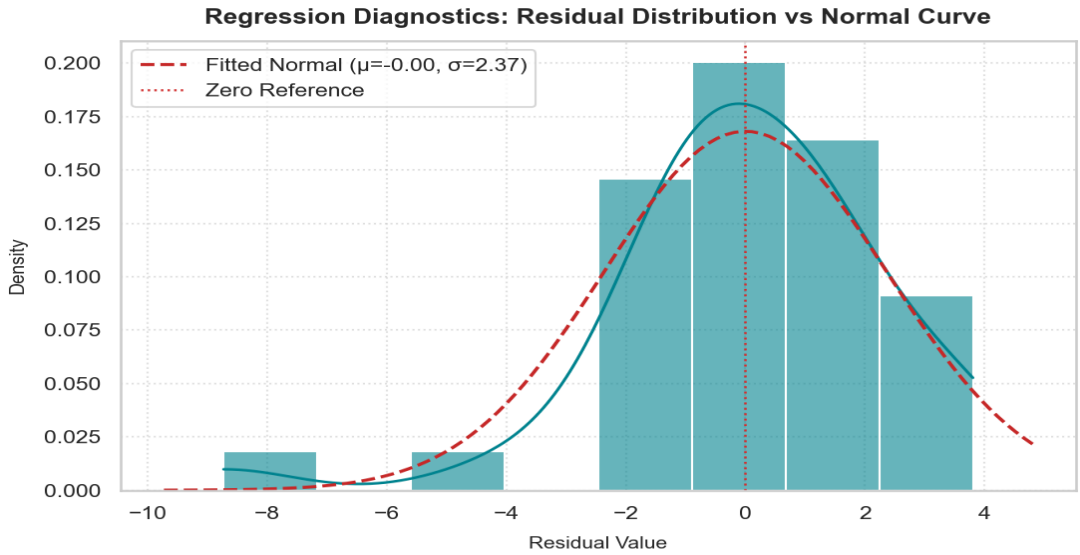


Figure 12: Empirical Distribution of Regression Residuals with Fitted Normal Curve

Histogram and KDE density of multiple regression residuals alongside fitted normal density curve (red dashed line).

9.3 Heteroskedasticity (Breusch-Pagan) and Serial Correlation (Durbin-Watson)

The Breusch-Pagan Lagrange Multiplier test statistic is LM = 8.6985 (p = 0.1217). Failing to reject H■ indicates no statistically significant heteroskedasticity at α = 0.05, though small-sample caution remains necessary. The Durbin-Watson statistic is 1.7435, indicating no severe first-order serial correlation.

9.5 Multicollinearity Assessment (Variance Inflation Factors)

Variance Inflation Factors (VIF) measure variance expansion resulting from inter-predictor collinearity. As detailed in Table 13 and Figure 13, Exports exhibits a VIF of 5.70, exceeding the moderate threshold of 5.0 due to its strong correlation with FDI (r = 0.77) and Capital Formation (r = 0.79). All other regressors remain below 5.0.

Independent Predictor	Variance Inflation Factor (VIF)	Tolerance (1/VIF)	Screening Status	Econometric Interpretation
Inflation (CPI)	1.1245	0.8893	Low Multicollinearity (VIF ≤ 5)	Acceptable collinearity profile
FDI Net Inflows	2.6866	0.3722	Low Multicollinearity (VIF ≤ 5)	Acceptable collinearity profile
Exports	5.6989	0.1755	Moderate Concern (VIF > 5)	Strong collinearity with FDI and Capital Formation
Gross Capital Formation	3.3785	0.2960	Low Multicollinearity (VIF ≤ 5)	Acceptable collinearity profile
Unemployment	1.7145	0.5833	Low Multicollinearity (VIF ≤ 5)	Acceptable collinearity profile

Rule of thumb: VIF < 5 indicates low collinearity; 5 ≤ VIF < 10 indicates moderate concern; VIF ≥ 10 denotes severe multicollinearity.

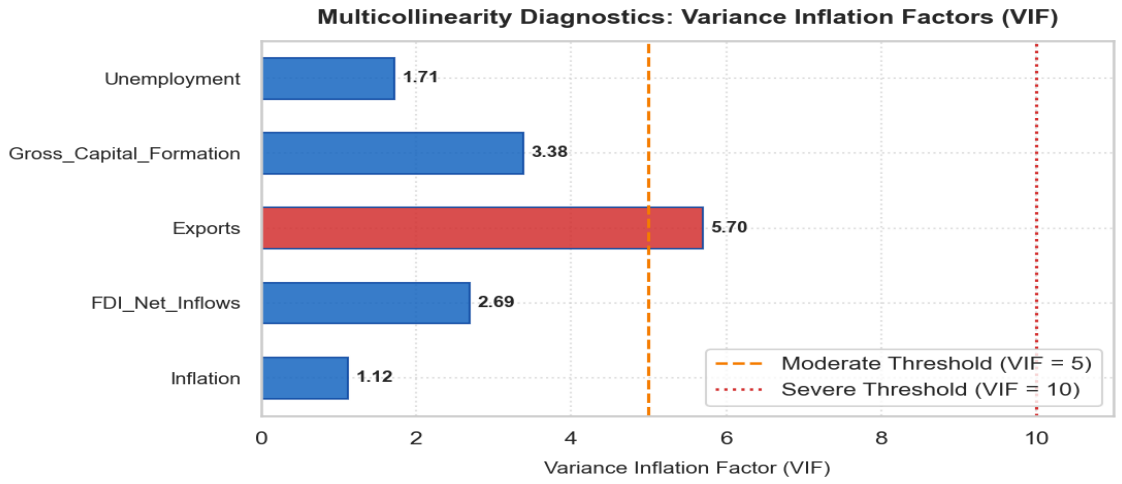


Figure 13: Variance Inflation Factors across Explanatory Variables

Bar chart illustrates VIF magnitudes. Orange dashed line marks moderate threshold (VIF = 5.0); red dotted line marks severe threshold (VIF = 10.0).

9.6 Cook's Distance and Influential Observations

Cook's Distance measures aggregate parameter displacement when an observation is omitted. Using the dynamic threshold 4/n ≈ 0.1143, exactly two points exceed the threshold: Year 1991 (D = 0.1786) and Year 2020 (D = 0.6139). Figure 14 illustrates the stem plot. These observations correspond to verified historical crises and are preserved in primary estimation.

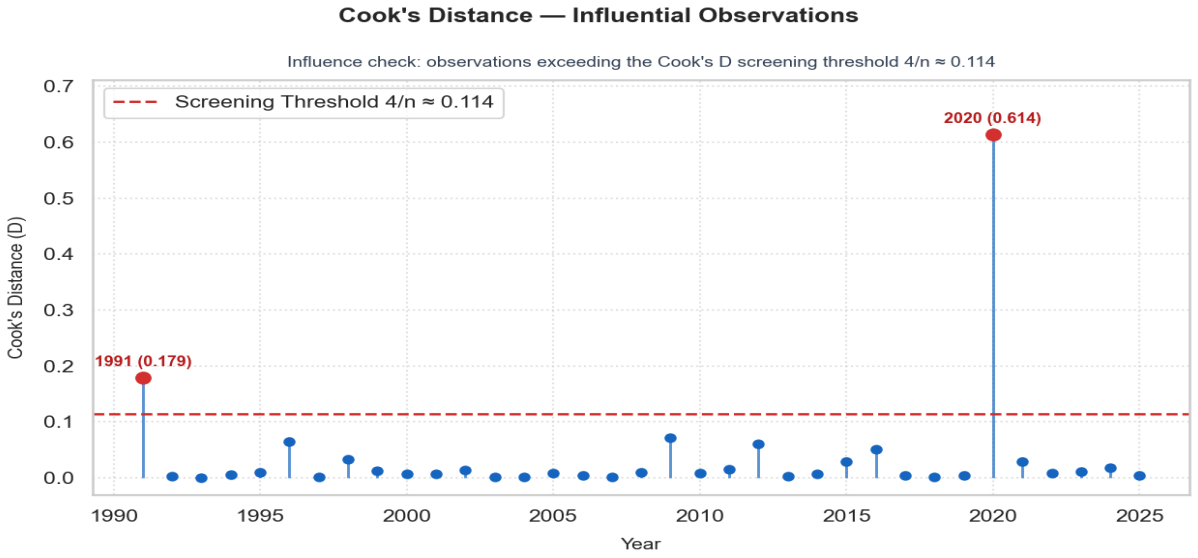


Figure 14: Cook's Distance Plot for Multiple Regression Model (India, 1991–2025)

Stem plot of Cook's D by year. Horizontal red dashed line marks the 4/n screening threshold (0.1143). Years 1991 and 2020 exceed threshold.

Table 14: Comprehensive Econometric Diagnostics Summary

Diagnostic Test	Null Hypothesis (H■)	Test Statistic	p-value / Threshold	Decision (α=0.05)	Econometric Assessment
Durbin-Watson Autocorrelation	No first-order serial correlation	1.7435	Threshold ~ 2.0	Fail to reject H■	No strong evidence of first-order serial correlation (DW = 1.744)
Breusch-Pagan Heteroskedasticity	Constant error variance (homoscedastic)	8.6985	0.1217	Fail to reject H■	No statistically significant evidence of heteroskedasticity (p = 0.1217)
Shapiro-Wilk Residual Normality	Error terms are normally distributed	0.9006	0.0041	Reject H■	Residuals deviate from normality (p = 0.0041); motivates HC3 robust standard errors
Cook's Distance Influence Check	No disproportionately influential points	Max D = 0.6139 (2020)	Threshold 4/n = 0.114	2 Exceeding Points	Years 1991 (0.179) and 2020 (0.614) flagged; reflect genuine historical crises
Variance Inflation Factor (VIF)	No severe linear multicollinearity	Max VIF = 5.6989 (Exports)	Threshold = 5.0 / 10.0	Moderate Concern	Exports exhibits moderate collinearity with FDI and Capital Formation

Comprehensive diagnostic evaluation based on verified statsmodels and scipy outputs.

Table 15: Formal Statistical Hypothesis Testing Summary (α = 0.05)

Hypothesis ID	Research Focus	Null Hypothesis (H■)	Alternative (H■)	Test Statistic & p-val	Decision (α=0.05)	Formal Research Conclusion
H■: Capital Formation	Simple OLS Slope	β_GCF = 0 (No linear effect)	β_GCF ≠ 0 (Linear effect)	t = 1.927, p = 0.0626	Fail to reject H■	Marginally non-significant linear association in bivariate simple regression.
H■: Model Joint Fit	Multiple Regression F-test	β■=β■=β■=β■=β■=0	At least one β■ ≠ 0	F = 2.300, p = 0.0707	Fail to reject H■	Joint model does not explain significant variance at α = 0.05.
H■: Robust Predictors	Individual HC3 Slopes	β■ = 0 under HC3 SE	β■ ≠ 0 under HC3 SE	All HC3 p > 0.13	Fail to reject H■	No single macroeconomic determinant remains significant under robust inference.
H■: Residual Normality	Error Distribution	Residuals are normal	Residuals are non-normal	W = 0.901, p = 0.0041	Reject H■	Residual non-normality confirmed; validates necessity of robust estimation.
H■: Homoscedasticity	Constant Error Variance	Var(ε|X) = σ²	Var(ε|X) ≠ σ²	LM = 8.698, p = 0.1217	Fail to reject H■	No significant heteroskedasticity detected by Breusch-Pagan test.

All hypothesis decisions evaluated strictly at the conventional two-tailed α = 0.05 significance level.

10. Discussion of Results

10.1 Descriptive and Distributional Findings: India's post-1991 macroeconomic performance illustrates strong expansion coupled with structural vulnerability to exogenous shocks. The divergence between parametric (Pearson) and non-parametric (Spearman) correlations highlights the impact of extreme points like 2020. Gross Capital Formation (rho = 0.4347, p = 0.0090) and Unemployment (rho = -0.3754, p = 0.0264) demonstrate highly significant monotonic relationships with growth that are obscured in linear space.

10.4 Multiple Regression and Robust Inference: The multivariate results underscore the necessity of econometric scrutiny. Under naive OLS estimation, FDI and Gross Capital Formation appear statistically significant. However, Davidson-MacKinnon HC3 robust covariance adjustments expand standard errors by over 38%–41%, rendering all predictors statistically non-significant. This divergence demonstrates how leverage points and non-Gaussian error terms can create spurious significance in small macroeconomic samples.

10.6 Methodological Limitations:

1. **Sample Size:** Annual data yields 35 observations, limiting statistical power and degrees of freedom (df = 29 in multiple regression).
2. **Observational Design:** Regression estimates conditional correlations, not structural causal impacts.
3. **Omitted Variable Bias:** Essential macroeconomic factors—including monetary policy rates, global oil prices, agricultural output, and fiscal deficits—are omitted from this baseline specification.
4. **Time-Series Non-Stationarity:** Static OLS assumes stationary series; integrated series can exhibit spurious regression tendencies.

11. Conclusion and Recommendations

11.1 Conclusion: Analyzing World Bank WDI data for India across 1991–2025 demonstrates that annual GDP growth cannot be adequately modeled as a simple static linear combination of standard macroeconomic aggregates. While domestic investment and labor market dynamics correlate significantly with growth ranks, the multiple linear regression model yields an overall non-significant fit at $\alpha = 0.05$ ($R^2 = 0.2840$, $p = 0.0707$). Robust standard errors and diagnostic checks reveal that apparent OLS significance is fragile and sensitive to historical shocks.

11.2 Recommendations for Advanced Statistical Modeling:
1. **Dynamic Cointegration Modeling:** Future research should employ Autoregressive Distributed Lag (ARDL) and Vector Error Correction Models (VECM) to test for cointegration and separate short-run fluctuations from long-run equilibrium relationships.
2. **Endogenous Structural Break Testing:** Formal testing for endogenous breaks (e.g., Bai-Perron) around 2008 and 2020 would prevent parameter distortion from major macroeconomic shocks.
3. **Quarterly Data Frequency:** Transitioning to quarterly macroeconomic accounts would increase sample size and statistical power.
4. **Expanded Conditioning Information:** Incorporating exchange rate volatility, policy repo rates, and fiscal balances would mitigate omitted variable bias.

11.3 Policy Implications: Policy initiatives seeking to stimulate long-term economic expansion must recognize that capital formation and export openness operate through complex structural channels rather than immediate linear impacts. Maintaining a stable macroeconomic environment with moderate inflation, targeted infrastructure investment, and proactive labor absorption remains foundational.

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Appendix: Comprehensive Statistical Documentation

This appendix compiles complete supporting numerical documentation, full time-series records, and mathematical formulas.

Appendix A: Complete Annual Macroeconomic Dataset for India (1991–2025, n = 35)

Year	GDP Growth (%)	Inflation (%)	FDI Inflows (%)	Exports (%)	Capital Form (%)	Unemployment (%)
1991	1.06	13.87	0.03	8.49	25.14	7.64
1992	5.48	11.79	0.10	8.84	27.44	7.65
1993	4.75	6.33	0.20	9.83	22.72	7.66
1994	6.66	10.25	0.30	9.89	26.31	7.59
1995	7.57	10.22	0.59	10.84	29.15	7.57
1996	7.55	8.98	0.62	10.39	22.76	7.54
1997	4.05	7.16	0.86	10.69	25.69	7.57
1998	6.18	13.23	0.63	11.02	24.98	7.60
1999	8.85	4.67	0.47	11.45	30.96	7.58
2000	3.84	4.01	0.77	13.00	25.68	7.59
2001	4.82	3.78	1.06	12.56	29.91	7.61
2002	3.80	4.30	1.01	14.26	30.25	7.68
2003	7.86	3.81	0.61	14.95	30.84	7.63
2004	7.92	3.77	0.77	17.86	35.10	7.60
2005	7.92	4.25	0.89	19.61	37.43	7.55
2006	8.06	5.80	2.13	21.27	39.00	7.55
2007	7.66	6.37	2.07	20.80	41.95	7.56
2008	3.09	8.35	3.62	24.10	38.42	7.63
2009	7.86	10.88	2.65	20.40	39.26	7.64
2010	8.50	11.99	1.64	22.40	39.79	7.63
2011	5.24	8.91	2.00	24.54	39.59	7.61
2012	5.46	9.48	1.31	24.53	38.35	7.64
2013	6.39	10.02	1.52	25.43	34.02	7.68
2014	7.41	6.67	1.70	22.97	34.27	7.66
2015	8.00	4.91	2.09	19.81	32.12	7.63
2016	8.26	4.95	1.94	19.16	30.17	7.61
2017	6.80	3.33	1.51	18.79	30.98	7.63
2018	6.45	3.94	1.56	19.93	32.34	7.65
2019	3.87	3.73	1.78	18.66	30.10	6.51
2020	-5.78	6.62	2.41	18.68	28.92	7.86
2021	9.69	5.13	1.41	21.40	32.12	6.38
2022	7.61	6.70	1.54	24.04	34.41	4.82
2023	7.21	5.65	0.80	22.35	34.50	4.17
2024	7.10	4.95	0.72	22.06	34.35	4.17
2025	7.57	2.40	1.03	22.26	34.60	4.22

Complete non-truncated dataset covering 35 annual observations extracted from World Bank WDI.

Appendix D: Full Correlation Matrices (Pearson r above diagonal, Spearman rho below diagonal)

Variable	GDP Growth	Inflation (CPI)	FDI Net Inflows	Exports	Gross Capital Formation	Unemployment
GDP Growth	1.000	-0.141	-0.118	0.193	0.318	-0.210
Inflation (CPI)	-0.133	1.000	-0.096	-0.221	-0.096	0.287
FDI Net Inflows	0.114	-0.097	1.000	0.697	0.626	0.107
Exports	0.231	-0.040	0.683	1.000	0.825	-0.351
Gross Capital Formation	0.431	-0.072	0.615	0.825	1.000	-0.156
Unemployment	-0.375	0.232	0.110	-0.083	-0.140	1.000

Above diagonal values denote Pearson r; below diagonal values denote Spearman rank rho.

Appendix I: Econometric Formulas, Units, and Estimation Notes

Sample Variance (Bessel's Correction): $s^2 = \Sigma (X■ - X■)^2 / (n - 1)$

Sample Standard Deviation: s = √[Σ (X■ - X■)² / (n - 1)]

Pearson Linear Correlation: r = Σ (X■ - X■)(Y■ - ■) / [√Σ(X■ - X■)² √Σ(Y■ - ■)²]

Spearman Rank Correlation: rho = 1 - [6 Σ d■² / (n(n² - 1))]

Ordinary Least Squares (OLS): β■ = (X'X)■¹ X'y, with Cov(β■) = s² (X'X)■¹

Davidson-MacKinnon HC3 Robust Covariance: Cov_HC3(β■) = (X'X)■¹ [Σ (e■ / (1 - h■■))² x■ x■'] (X'X)■¹ where h■■ = x■'(X'X)■¹x■ is leverage.

Durbin-Watson Statistic: DW = Σ (e■ - e■■■)² / Σ e■²

Variance Inflation Factor: VIF■ = 1 / (1 - R■²)

Cook's Distance: D■ = [e■² / ((k + 1) s²)] · [h■■ / (1 - h■■)²]

Shapiro-Wilk Test Statistic: W = (Σ a■ x■■■)² / Σ (x■ - x■)²